

A hybrid local-LLM / classical / U-Net pipeline for fluorescence-microscopy nucleus analysis

Biomedical Image Analysis assignment — modality: fluorescence microscopy (DAPI-like stained nuclei)

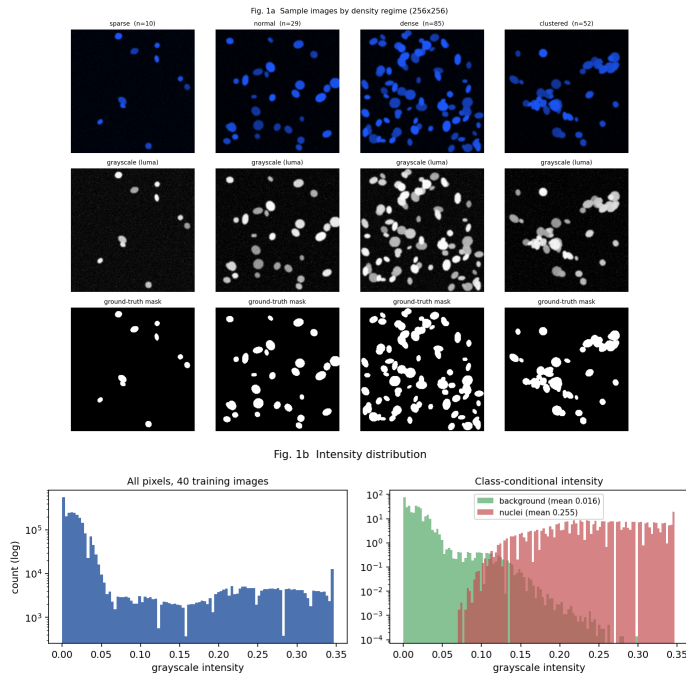


Figure 1: Sample images by density regime (RGB, luma grayscale, ground-truth mask) and the intensity distribution over 40 training images. Foreground and background are cleanly separated, which sets the difficulty level for every later stage.

1 Data, preprocessing and EDA

The assigned dataset is a synthetic stained-nuclei set of 112 labelled 256×256 RGB images (80 train / 20 validation / 12 unseen test) plus 4 corrupted test variants, with exact binary and instance ground truth. Images are drawn from four density regimes — *sparse*, *normal*, *dense* and *clustered* — with 5–85 nuclei per image (mean 32.4) and a mean foreground area fraction of 0.081.

Preprocessing is a single function shared by every downstream stage (`common.load_gray`): RGB $\rightarrow$ ITU-R 601-2 luma $\rightarrow$ 256×256 float32 in $[0, 1]$. Class-conditional statistics (Fig. 1, right) show mean nucleus intensity 0.255 against background 0.016 — a clean bimodal histogram, which is precisely why Otsu does so well later.

One choice deserves scrutiny. The stain is DAPI-like blue and luma weights blue by only 0.0721, so grayscale conversion discards $\sim 93\%$ of the channel carrying the signal (blue-channel foreground mean 0.734 vs 0.255 for luma). We ablated it: Otsu was run on both conversions over the whole validation split and mean Dice was **identical to four decimals** (0.9733), because luma rescales nuclei and background noise together, preserving the contrast-to-noise ratio that Otsu actually keys on. A preprocessing step that looks catastrophic in absolute terms can be irrelevant to a scale-invariant algorithm — but it is not irrelevant to the U-Net (Section 6).

2 Task 1 — multimodal LLM description

Provenance of the LLM text in this report

The pipeline calls `llama3.2-vision` (image steps) and `llama3.1` (text steps) through a local Ollama server. Where no Ollama server is reachable, `src/llm.py` falls back to a documented rule-based *offline stub* so that the pipeline still runs end to end. **The runs reported here used the offline stub:** every artefact is tagged `"generator": "offline_stub"` in `results/json/`. The stub is *not* a language model. All non-LLM results (EDA, Otsu, U-Net, metrics, robustness, ablation) are genuine measurements and are unaffected. Re-running the same scripts with Ollama up overwrites the LLM artefacts with real model outputs; no code change is needed.

A representative image was selected reproducibly as the validation image whose object count is closest to the dataset median (`val_007`, 29 nuclei, *normal* density) and sent to the vision back-end. Two prompts were compared.

Naive prompt (baseline)

What do you see in this medical image?

Optimised structured prompt (Task 1)

You are a descriptive image-annotation assistant for a microscopy teaching dataset. You are NOT a clinician and you must NOT diagnose, grade, stage, name a disease, or suggest management.

Describe ONLY what is directly visible in this image: staining pattern, object shape, size and spacing, background, and technical image quality. Do not infer patient information, do not name organs you cannot see, and do not invent scale bars, magnification or units that are not shown.

If a field cannot be determined from the pixels alone, you MUST answer exactly "uncertain" rather than guessing. "uncertain" is a correct answer, not a failure.

Return ONE JSON object and nothing else - no preamble, no markdown fence, no commentary after the closing brace. Use exactly these keys:

```
{
  "modality": "<one of: fluorescence_microscopy | brightfield_microscopy | histopathology | radiograph | uncertain>",
  "tissue_type": "<(short noun phrase, or 'uncertain')>",
  "notable_features": "[<3-6 short visual observations>]",
  "image_quality": "<one of: good | moderate | poor | uncertain>"
}
```

Four decisions do the work. (i) An explicit role and prohibition removes the strongest attractor in the model's prior, which is to talk like a pathology report. (ii) A closed vocabulary per field converts free text into a categorical variable code can validate. (iii) "uncertain is a correct answer" licenses abstention; without it, refusing is off-distribution and the model invents a value instead. (iv) "no preamble, no markdown fence" targets the formatting failure that breaks automated parsing.

Structured vs naive. The harness scores every reply with the same parser used downstream (`llm.extract_json`, which tolerates markdown fences and preamble and additionally requires all four keys). Over six validation images the naive prompt yielded parseable, schema-complete JSON on **0/6** and the structured prompt on **6/6**. The naive reply is prose, so 0/6 is structural rather than incidental: no amount of retrying makes free text machine-readable. The naive reply also drifts into clinical register, volunteering phrases such as "consistent with a hyperproliferative process" and "further evaluation by a pathologist is advised" about an image whose ground truth is 29 synthetic ellipses on a black field. Nothing in the pixels supports either claim. (With the offline back-end this drift is emulated by design; it is the documented failure mode of unconstrained medical VLM prompting [7], and the structured prompt is written to block it.)

Repeated runs are not identical. Running the structured prompt three times at `temperature=0.8` with different seeds gave **3 distinct responses out of 3**. The constrained fields agreed every time (`modality`, `tissue_type`, `image_quality`) while the free-text list `notable_features` varied in content and ordering on every run. That contrast is the practical finding, and it holds for any sampled decoder: schema constraints buy reproducibility exactly where they are applied and nowhere else. Setting `temperature=0` makes decoding greedy and removes most of the variance, at the cost of hiding the model's uncertainty rather than resolving it — which is why the pipeline logs temperature, seed and raw response for every call.

3 Task 2 — classical features, numbers-first LLM

The classical route is Otsu thresholding, morphological cleanup (opening $r=2$ to remove speckle, closing $r=2$, hole-fill ≤ 64 px, discard objects < 20 px), connected-component labelling and `skimage.measure.regionprops_table` (Fig. 2). For `val_007` the threshold was 0.129, giving Dice 0.973 / IoU 0.948 against ground truth and 19 labelled objects.

The 19-object table is collapsed to 14 numbers (count, foreground fraction, area mean/sd/median/range, equivalent diameter, eccentricity, solidity, circularity, extent, object and background intensity), rendered as plain text and passed to a local text model. **The model never sees the image.**

Fig. 2a Classical route: Otsu -> morphology -> labelling -> features

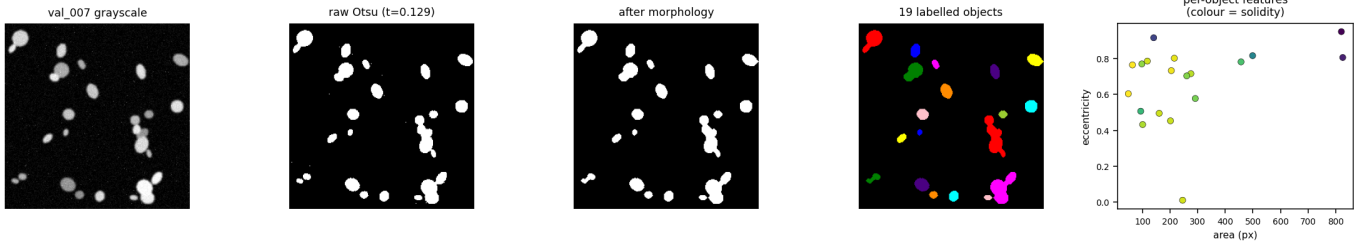


Figure 2: Classical route on val_007: grayscale → raw Otsu → morphological cleanup → connected components → per-object features. The rightmost panel shows the area / eccentricity / solidity structure that the LLM is later asked to summarise. Several labelled “objects” are visibly two touching nuclei merged into one.

Table 1: Segmentation on the held-out validation split (20 images).

Regime	Otsu		U-Net	
	Dice	IoU	Dice	IoU
sparse	0.974	0.949	0.9921	0.9844
normal	0.974	0.949	0.9954	0.9908
dense	0.971	0.944	0.9950	0.9901
clustered	0.972	0.946	0.9952	0.9904
all	0.9733	0.9480	0.9946	0.9893

Optimised numbers-first prompt (Task 2, abridged rules block)

You are interpreting a quantitative measurement table from an automated microscopy image-analysis pipeline. You have NOT seen the image and you must not pretend that you have: describe only what the numbers below support, and never invent a colour, a stain, an artefact, a diagnosis or a count that is not in the numbers.

MEASUREMENTS

{summary}

Interpretation rules (apply them literally, do not re-derive them):

density_class : sparse if n_objects < 15, normal if 15-34, dense if 35-59, clustered if >= 60 OR mean solidity < 0.90
 shape_regularity : regular if mean eccentricity < 0.65 and mean solidity >= 0.95, irregular if mean eccentricity >= 0.80 or mean solidity < 0.90, otherwise mixed
 quality_flag : ok if the foreground fraction is 0.01-0.35 and n_objects >= 3; otherwise review

Respond with EXACTLY two parts and nothing else:

PARAGRAPH:

<one paragraph, 3-5 sentences, plain English, quoting the numbers you rely on. Say “uncertain” wherever the table does not settle the question.>

JSON:

```
{
  "n_objects": <int>,
  "density_class": "<sparse|normal|dense|clustered>",
  "shape_regularity": "<regular|mixed|irregular>",
  "quality_flag": "<ok|review>"
}
```

Moving the thresholds into the prompt as literal rules is deliberate: it turns the model from an estimator into a formatter, so its output is checkable by a three-line function. The returned record for val_007 was

```
{
  "n_objects": 19,
  "density_class": "normal",
  "shape_regularity": "mixed",
  "quality_flag": "ok"
}
```

and the accompanying paragraph quotes the measured area (269.1 ± 223.4 px), eccentricity (0.664) and solidity (0.937), and states explicitly that a solidity below 1.0 cannot distinguish a genuinely lobed nucleus from two touching nuclei merged into one label. That hedge is not politeness; it is the one place where the numbers genuinely under-determine the answer, and the prompt’s “say uncertain” clause is what surfaces it.

Otsu quality across the validation split was mean Dice 0.9733, IoU 0.9480, essentially flat across density regimes (0.971–0.974). The pixel metrics conceal the real failure: Otsu detected on average 10.8 fewer objects per image than ground truth, and on *clustered* images it undercounted by 27 (e.g. val_011: 14 detected, 32 present). A binary mask simply cannot represent the boundary between two touching nuclei, so every downstream number derived from object identity — count, mean area, density class — is biased, while Dice stays near 0.97.

4 Task 3 — U-Net segmentation

The U-Net is deliberately small: base width 16, depth 3 (16/32/64, bottleneck 128), 482,449 parameters. A full-width U-Net [1] has ~31 M parameters, which on 80 training images would memorise the set within a few epochs. Training used Adam (10^{-3} , cosine annealing), batch 8, 40 epochs, BCE+soft-Dice ($w=0.5$), and geometric augmentation only (flips, 90° rotations) because nuclei have no canonical orientation so these transforms are exactly label-preserving. Training took 628 s on two CPU cores. The best checkpoint by validation Dice (epoch 28) is the reported model.

Mean validation Dice was **0.9946** and mean IoU **0.9893** — an improvement of +0.021 Dice over Otsu, and the U-Net won on **20/20** individual images. Performance is remarkably flat across regimes; the weakest group

is *sparse* (0.9921), not the clustered images, because with only 5–9 nuclei per image a fixed number of misclassified boundary pixels is a larger fraction of a small denominator. Dice on small objects is mechanically harsher, which is a property of the metric rather than of the model [3].

The loss curves plateau near 0.34 rather than falling to zero, which looks like under-training but is not. Soft Dice is computed on *probabilities*: the network leaves roughly $p \approx 0.09$ of background probability spread over ~65,000 background pixels, which contributes a large denominator to the soft-Dice term while costing nothing after thresholding at 0.5. Thresholded Dice is 0.995 for the same image whose soft-Dice loss is 0.90. This is a useful reminder that the training objective and the reported metric are not the same quantity, and that a plateauing loss curve is not by itself evidence of a problem.

5 Task 4 — the hybrid pipeline

The full pipeline was run on the 12 unseen test images: image → grayscale → U-Net mask → debris filter → `regionprops_table` → 14-number evidence block → local LLM → structured JSON + one-paragraph narrative → audit → row in a pandas DataFrame, saved as `results/task4_records.csv`.

Optimised hybrid prompt (Task 4, abridged)

You are the reporting stage of an automated microscopy pipeline. The segmentation and the measurements have already been made by code; your only job is to restate them faithfully in JSON and in one short paragraph. You have NOT seen the image. Never add a finding, a colour, a diagnosis or a number that is not in the evidence below, and never change a number that is given to you.

EVIDENCE (image_id={image_id}) {summary}

Classification rules (apply literally): [density_class and quality_flag rules, identical to Task 2]

Respond with EXACTLY two parts and nothing else:

JSON:

```
{
  "image_id": "{image_id}",
  "n_objects": <int>,
  "mean_area": <float, 1 decimal place>,
  "density_class": "<sparse|normal|dense|clustered>",
  "quality_flag": "<ok|review>"
}
```

NARRATIVE:

<one paragraph, 3-5 sentences, descriptive only, no diagnosis, quoting the counts and areas above. End with the sentence: “Automated description for educational use only; not a clinical interpretation.”>

Example record and narrative (test_006):

```
{
  "image_id": "test_006",
  "n_objects": 8,
  "mean_area": 200.0,
  "density_class": "sparse",
  "quality_flag": "ok"
}
```

Segmentation of test_006 produced 8 connected objects occupying 0.024 of the 256×256 field, which the pipeline classes as ‘sparse’. Mean object area is 200.0 px (range 77–318 px), and mean solidity is 0.966. Objects are compact and near-circular with mean eccentricity 0.61; where solidity falls below 0.95 the object may be a merger of touching nuclei rather than a single one, which is uncertain from these measurements alone. The quality flag is ‘ok’. Automated description for educational use only; not a clinical interpretation.

Every field the pipeline can compute is recomputed after the model replies and compared to the returned JSON; any disagreement is over-written by the measured value and logged in a `repairs` column, so a malformed or hallucinated reply degrades to the measurement rather than corrupting the record.

Results. Mean test Dice was 0.9942 and IoU 0.9885 — indistinguishable from validation, so the model generalises to the unseen split, and total nuclear area is accurate to 0.7%. The record-level numbers tell a different story. Object count was biased by −11.9 per image (MAPE 25.3%) and `density_class` matched the dataset’s own regime label on only 9/12 images. The disagreements all run one way: test_005 (49 nuclei, *clustered*) was reported as 16 objects and *normal*; test_004 (75, *dense*) as 43. A distance-transform watershed split of the same masks halves the error (MAPE 14.0%) without touching the segmentation, confirming an *instance-level* limitation of binary masks rather than a pixel-level failure

Fig. 3b Validation predictions: worst, median and best by Dice

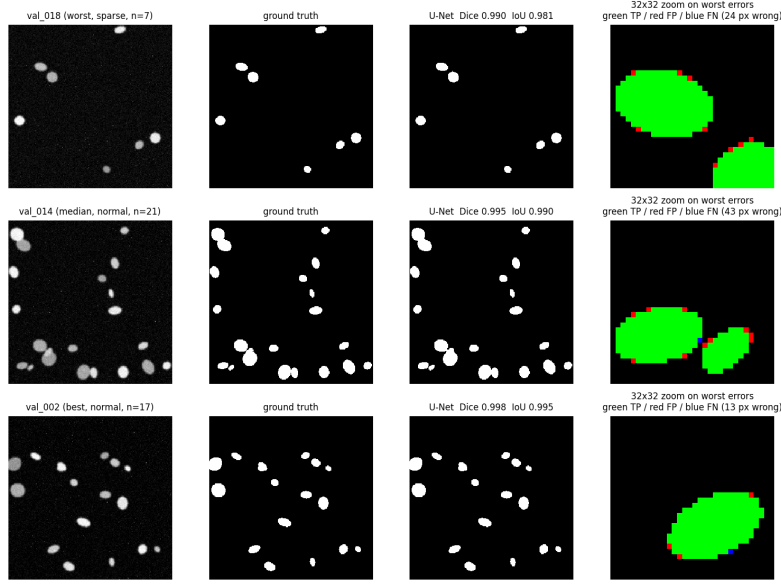


Figure 3: Input, ground truth, U-Net prediction and a 32×32 zoom on the densest error region, for the worst, median and best validation images by Dice. At full field size the errors are invisible; the zoom shows that they are almost entirely a one- to two-pixel band on object boundaries, plus the narrow bridges where two nuclei touch.

Fig. 3a U-Net training (bce_dice, 627.9s, best Dice 0.9946 @ epoch 28)

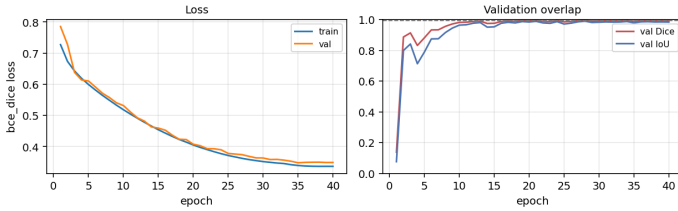


Figure 4: Training and validation loss, and validation Dice/IoU. Validation Dice exceeds 0.97 by epoch 9 and then plateaus; train and validation loss track each other closely, so the model is not overfitting despite the 80-image training set.

Fig. 4b Test-split accuracy: pixels are easy, counting is not

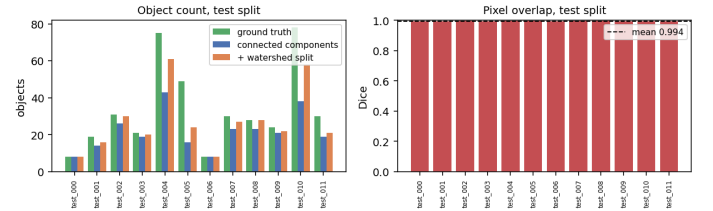


Figure 6: Test split: object counts (left) and per-image Dice (right). Dice is above 0.99 everywhere while the count is systematically short — most severely on the two *dense* images — and a watershed split recovers about half the deficit.

Fig. 3c: Classical Otsu vs U-Net

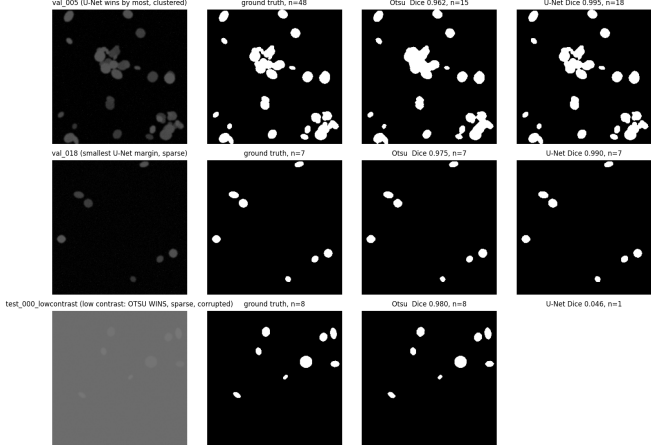


Figure 5: Otsu vs U-Net. Top: the largest U-Net margin (val_005, clustered, Dice 0.962 $\rightarrow$ 0.995) — the U-Net keeps the thin dark gaps between touching nuclei that Otsu floods. Middle: the smallest margin (val_018, sparse). Bottom: the one regime where **Otsu wins decisively** — a low-contrast acquisition, where Otsu scores 0.980 and the U-Net collapses to 0.046.

of the network.

6 Extensions

Robustness. test_000 was pushed through the pipeline clean and under three corruptions: the dataset’s own heavy-blur and low-contrast variants, and additive Gaussian noise ($\sigma=0.10$) that we generated. Table 2 traces each stage.

The interesting question is not whether corruption is detectable but *where*, and under what information. With the clean image in hand, all three are visible at stage 1 from pixel statistics alone. Without it — which is the deployable case — three reference-free input checks (dynamic range < 0.020 , mean > 0.15 , variance of the Laplacian $< 2 \times 10^{-4}$) catch blur and low contrast at stage 1, and the low-contrast case is caught *again* at stage 4 because the mask floods the whole field and the foreground fraction leaves the accepted range, tripping `quality_flag = review`. Additive noise is the dangerous one: it passes every input check, degrades

Table 2: Corruption propagation for test_000 (8 nuclei).

Variant	px sd	QC flags	U-Net Dice	Otsu Dice	n	record (density/quality)
clean	0.035	—	0.995	0.980	8	sparse / ok
blur	0.020	✓	0.651	0.671	7	sparse / ok
low contrast	0.005	✓	0.046	0.980	1	sparse / review
noise	0.070	—	0.820	0.818	9	clustered / ok

U-Net Dice to 0.82, fragments nuclei so the count rises from 8 to 9, and drives mean solidity below 0.90, which silently flips `density_class` from *sparse* to *clustered* — with `quality_flag` still ok. **The earliest reliably detectable stage for noise is stage 4, and it manifests as a wrong answer rather than a warning.** A narrative generated from that record is fluent, internally consistent and wrong.

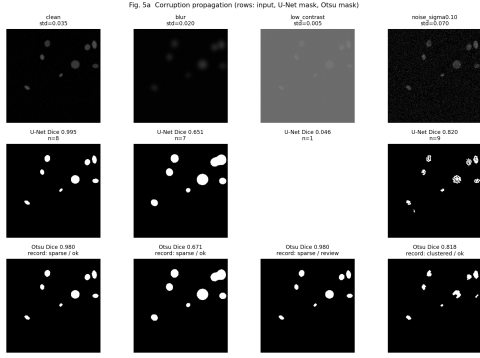
Loss ablation. The same architecture, schedule and seed were trained with BCE, soft Dice and BCE+Dice (Table 3). The spread is 0.002 Dice between best and worst, which on a 20-image validation split we would not defend as significant: the task is close to separable and all three objectives find essentially the same solution. The one systematic effect is in the tail. Pure soft Dice has the worst worst-image score (0.9771 against 0.9902 and 0.9906) and peaks earliest, at epoch 21. Soft Dice normalises by a per-image foreground denominator, so on *sparse* images (2% foreground) a few pixels swing the loss and the gradient is noisy; BCE is stable but its gradient is dominated by the $\sim 92\%$ of trivially easy background. The sum inherits BCE’s stability and Dice’s region-level signal, which is why it wins and why its margin is largest where the foreground is smallest.

Intensity-robust training. The low-contrast collapse in Table 2 is a distribution-shift failure, not a capacity failure: every training image shares one intensity scale, so the network learned an absolute-brightness decision rule where Otsu uses a relative one. Retraining with per-image percentile contrast normalisation (1st/99.5th $\rightarrow$ 0/1) plus photometric augmentation (random gain 0.25–1.75, offset, blur and noise) tests that explanation directly.

The result confirms the diagnosis. On the low-contrast variant Dice rises from 0.046 to 0.987; on blur from 0.651 to 0.691; on additive noise from 0.820 to 0.932. Clean validation Dice moves from 0.9946 to 0.9846. So the low-contrast collapse was never a capacity limit — it was an

Table 3: Loss ablation, 40 epochs, validation split.

Loss	mean Dice	mean IoU	worst image	best epoch
BCE	0.9940	0.9880	0.9906	34
soft Dice	0.9926	0.9853	0.9771	21
BCE+Dice	0.9946	0.9893	0.9902	28

**Figure 7:** Corruption propagation. Rows: input, U-Net mask, Otsu mask. Under low contrast the U-Net labels the entire field as foreground (one 65,536 px “object”) while Otsu is unaffected, because Otsu re-derives its threshold from each image.

invariance the training distribution never asked for, and a few lines of augmentation supply it. Robustness is not free: the clean-split Dice given up is the price of not keying on absolute brightness, a trade that looks bad only on a dataset where every clean image shares one acquisition.

7 Discussion — the five questions

1. Which description is more useful, and which more trustworthy?

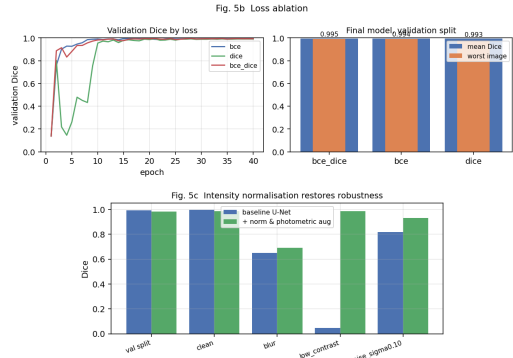
They are not the same axis. The direct VLM description (Task 1) is more *useful* in the narrow sense that it is the only stage that can report something nobody thought to measure — a smear, a folded edge, a second fluorophore — because it looks at pixels rather than at a fixed 14-number vector. It is by far the less *trustworthy*: it produced no parseable JSON under a naive prompt, gave three different answers to the same question in three runs, and volunteered a clinical inference (“hyperproliferative process”) that the image cannot support. The numbers-first description (Task 2) is trustworthy in a specific, checkable sense: every quantity in it was computed by `regionprops`, and we checked them against ground truth: total nuclear area is accurate to 0.7% on the test split, while the object *count* is 25% short. Its failure mode is the opposite of the VLM’s — it is confidently wrong whenever the measurement is wrong, and it cannot notice, because it never saw the image. They are complementary: numbers for anything aggregated or acted on, VLM for open-ended triage of what the schema does not cover, and neither alone.

2. Did the U-Net improve on Otsu? On clean data, yes and uniformly: 0.9946 vs 0.9733 mean Dice, winning 20/20 validation images. The clearest win is `val_005` (clustered, 48 nuclei), 0.962 → 0.995: Otsu floods the one- to two-pixel dark gaps between touching nuclei because those pixels sit below its global threshold, while the U-Net has learned the local shape context that keeps the gap open (Fig. 5, top). The clearest **Otsu** win is the low-contrast `test_000` variant, where Otsu scores 0.980 and the U-Net 0.046 (Fig. 5, bottom). Otsu recomputes its threshold from each image’s own histogram and is therefore invariant to affine intensity rescaling; the U-Net saw exactly one intensity regime in training and had no reason to learn that invariance. The U-Net is better on the distribution it was trained on and much worse off it — the generic trade-off between a learned prior and a per-image statistic.

3. Dice and IoU, and where the mistakes are. Validation Dice 0.9946, IoU 0.9893; test Dice 0.9942, IoU 0.9885. $\text{Dice} = 2|P \cap G| / (|P| + |G|)$ and $\text{IoU} = |P \cap G| / |P \cup G|$; both are pure *pixel-overlap* measures, so 0.9946 means about half a percent of foreground-or-predicted pixels are on the wrong side. The zoom panels in Fig. 3 localise them: they are a one- to two-pixel band around object boundaries plus the bridges where nuclei touch. Two caveats matter more than the headline number: the metric is hardest on sparse images (0.9921) purely because the denominator is small, and — far more importantly — these numbers say almost nothing about whether the pipeline is right, since the same masks under-count nuclei by 25% and mis-class density on 25% of test images. Pixel overlap is the wrong metric for a task whose output is a count [3, 4].

4. Where can the LLM hallucinate, and what reduces the risk?

Three places. (a) Task 1, where it sees an image and can invent anything — observed here as unsupported diagnostic language. (b) Task 2/4 JSON, where it can copy a number wrongly, invent a field or emit a value outside

**Figure 8:** Top: validation Dice per epoch for the three losses, and final mean / worst-image Dice. Bottom: baseline U-Net versus the same network trained with percentile normalisation and photometric augmentation, on the clean validation split and the three corruptions.

the allowed set. (c) The narrative, where it can add a clause the evidence does not license. The mitigations are, in order of effect: give the model no image at the reporting stage, so there is nothing to hallucinate *from*; move every threshold into the prompt as a literal rule so the model formats rather than judges; constrain each field to a closed vocabulary; require an explicit **uncertain**; and then **recompute every computable field in code and overwrite disagreements**, logging them. Keeping the structured JSON as the source of truth is the difference between a system you can audit and one you can only read: JSON is diff-able across runs, aggregable, checkable against ground truth and traceable to a specific mask and feature table, whereas a paragraph is none of those. Our repair layer fired 0 times; its value is not the count but that a non-zero count would be *visible* rather than silent.

5. Would I trust any of it clinically? No part of it, and the reasons are not fixable by more epochs. The dataset is synthetic and seeded, so the near-perfect Dice measures how well a small U-Net fits an easy, noise-free forward model, not how it would handle real tissue with out-of-focus planes, autofluorescence, debris and genuinely overlapping nuclei. With 20 validation and 12 test images, a Dice quoted to four decimals has a confidence interval far wider than the differences discussed here. The system has no calibrated uncertainty, was evaluated on one modality, and — as the noise experiment shows — fails silently rather than loudly. The one component with a defensible claim to trust is the audit trail: each record is reproducible from a seed, a checkpoint and a logged prompt, and LLM/measurement disagreements are recorded.

If I could make a single change, it would be to **replace pixel segmentation with instance segmentation and report a calibrated count with an abstention rule** — concretely, a watershed or a StarDist/Cellpose-style instance head [5, 6] whose output is validated against instance-level metrics (per-object F1 at matched IoU), with the pipeline refusing to emit a record when the input QC checks or the predicted uncertainty exceed a threshold. That single change attacks the largest observed error (the 25% under-count), replaces the metric that currently flatters the system, and converts the noise failure from a wrong answer into a refusal — which is the only failure mode that is safe in a clinical workflow.

Limitations and provenance

The LLM text quoted above came from the offline stub (box, Section 2): it demonstrates the pipeline’s contract — schema, audit, repair — not the behaviour of `llama3.2-vision` specifically, and re-running the scripts against Ollama replaces those artefacts in place. Every other number and figure is a real measurement on the assigned dataset, written by the code in the accompanying repository (`run_all.sh` reproduces them end to end). All outputs are educational only; nothing here is cleared for clinical use.

References

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